DATA STORYTELLING & STATISTICAL VALIDATION

ApexPlanet Data Analytics Internship — Task 4

Sales Analysis | Business Insights | Hypothesis Testing

1,000	■139,399,439.65	5,435	5
Dataset records	Total sales	Units sold	Product categories

Prepared from the supplied ApexPlanet sales dataset

1. Business Objective & Data Story

Objective: Turn the sales dataset into a concise business narrative and use a statistical test to check whether observed sales differences across categories are statistically meaningful.

Business questions

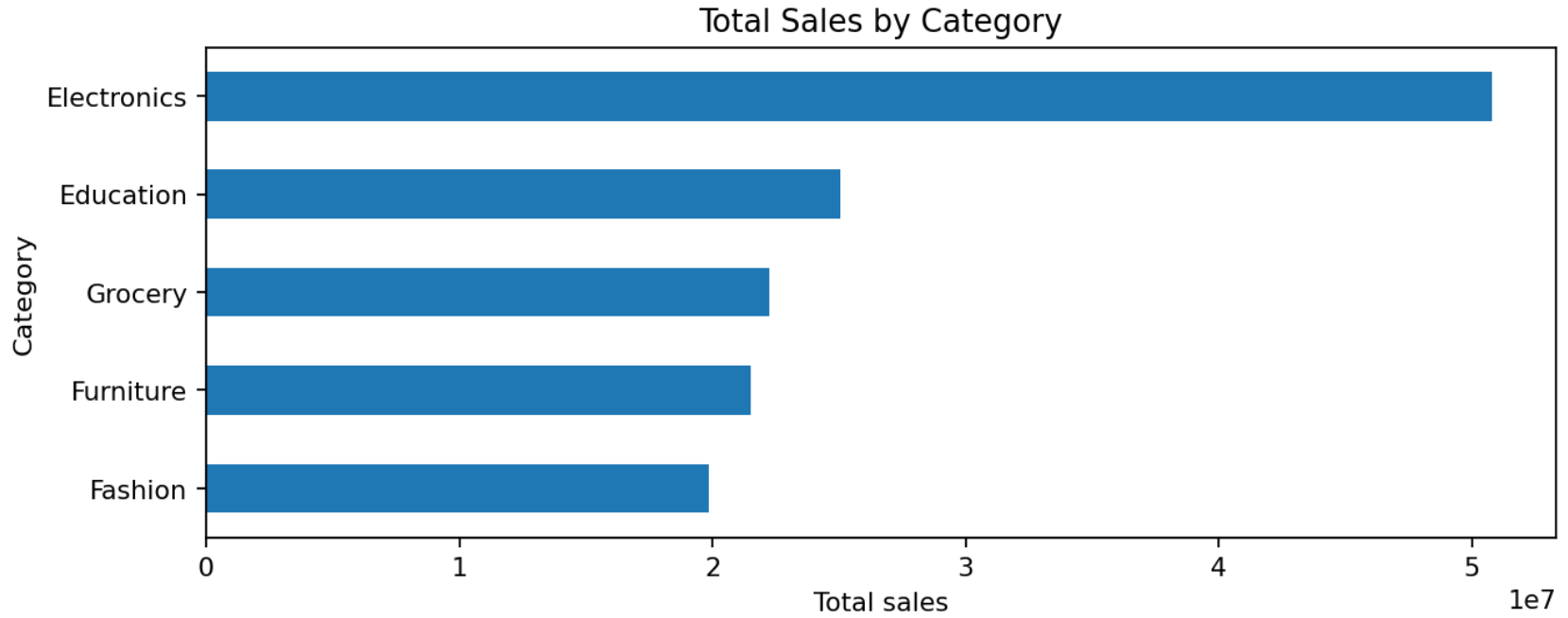
- Which product category contributes the most revenue?
- Which city generates the highest sales?
- How does revenue change month to month?
- Do average transaction sales differ significantly among categories?

Dataset snapshot

Metric	Value
Rows / records	1,000
Columns	12
Date coverage	01 Jan 2025 to 01 Jan 2026
Cities	8
Categories	5
Missing values in raw file	33 (Age: 20; City: 13)

2. Category Performance

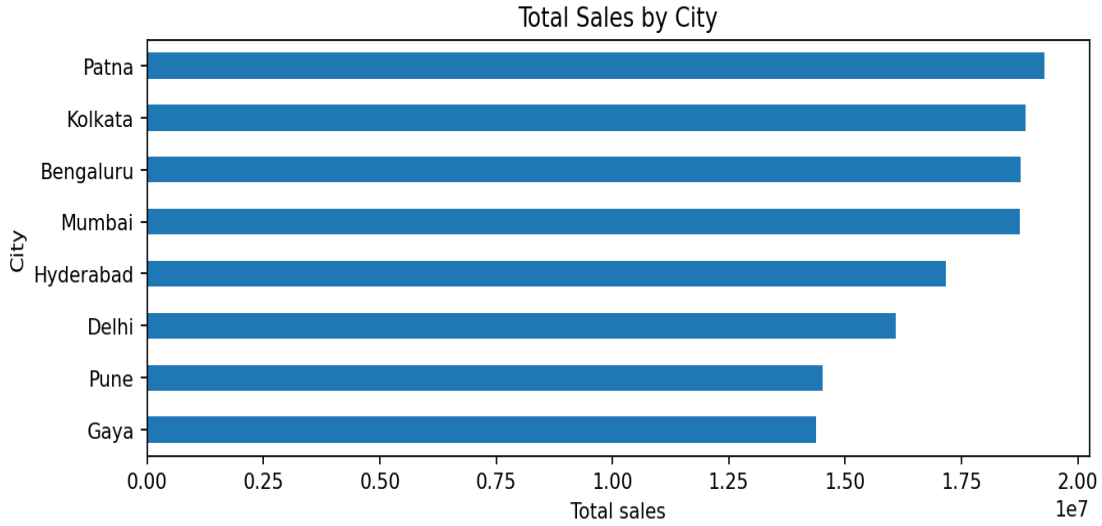
Key finding: Electronics is the largest revenue contributor, generating ■50,778,581.70 (36.4% of total sales).

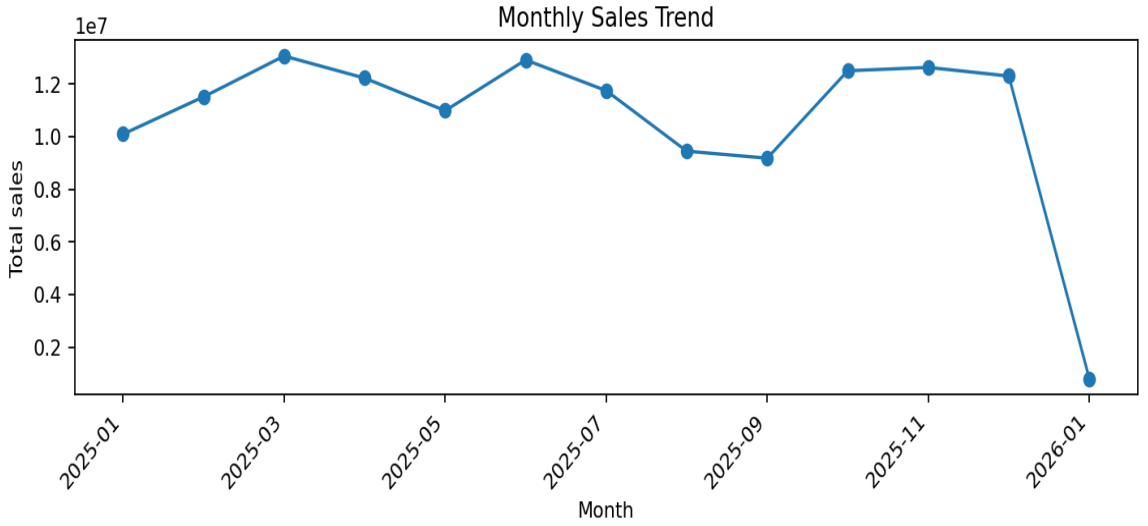


Category	Total sales	Share of sales	Average transaction
Electronics	■50,778,581.70	36.4%	■143,442.32
Education	■25,031,689.40	18.0%	■140,627.47
Grocery	■22,231,711.28	15.9%	■145,305.30
Furniture	■21,521,561.48	15.4%	■135,355.73
Fashion	■19,835,895.79	14.2%	■127,153.18

3. Geographic & Monthly Trends

Top city: Patna, with ■19,285,966.89 in sales. **Strongest month:** 2025-03, with ■13,059,899.94 in sales.





Business interpretation

- Compare the highest-revenue city with its customer base and product mix before reallocating budgets.
- Investigate monthly peaks for campaigns, seasonal demand, or large transactions.
- Use a rolling monthly dashboard to spot changes early; one high month alone does not prove a sustained trend.

4. Statistical Validation: Category Sales

Question: Are mean transaction sales different across the five product categories?

Null hypothesis (H■): All category mean transaction sales are equal.
Alternative hypothesis (H■): At least one category mean transaction sales differs.

Method: One-way ANOVA, using a 5% significance level ($\alpha = 0.05$).

Result	Value
F-statistic	0.7171
p-value	0.5803
Significance level (α)	0.05
Decision	Fail to reject H■
Conclusion	No statistically significant evidence of a difference in mean transaction sales across categories at the 5% level.

Interpretation: Electronics has the highest total revenue largely alongside the greatest number of records. The ANOVA result says the observed differences in average transaction sales are not statistically significant at $\alpha = 0.05$. This is not proof that the categories are identical; it means this test did not find sufficient evidence of a mean difference.

Note: This is a comparison of observed transaction sales by category, not an A/B experiment or proof of causal impact. ANOVA conclusions depend on assumptions such as independent observations and reasonably suitable within-group distributions.

5. Recommendations & Call to Action

Recommended actions

1. **Prioritize category diagnostics:** Review Electronics revenue and order volume together; total revenue alone can be influenced by the number of transactions.
2. **Investigate city-level opportunities:** Review product mix, customer segments, and marketing activity in Patna and other high-performing cities.
3. **Monitor monthly performance:** Track revenue, transaction count, average transaction value, and units sold in one dashboard.
4. **Improve data quality:** Resolve missing Age and City values using documented rules and review repeated Order_ID values before treating them as unique transactions.
5. **Validate before acting:** For stronger decisions, examine effect sizes, confidence intervals, and model assumptions; use controlled experiments when testing a new business intervention.

Closing story

The dataset records ■139,399,439.65 in total sales across 1,000 rows. Electronics contributes the largest share of revenue, while Patna is the highest-revenue city. Monthly results vary, so ongoing monitoring is useful. Statistical testing did not find a significant difference in mean transaction sales across categories at the 5% level; therefore, category-level decisions should combine revenue, volume, profitability, and further validation.

Prepared for ApexPlanet Data Analytics Internship — Task 4 | Source: user-provided sales dataset